



IDC 203: INTRODUCTION TO EARTH SCIENCES



Wind Erosion

- Wind erodes dry land much more effectively than it does moist land.
- As the wind erodes land it carries rock particles along with it, mostly sand, silt, and clay.
- Sand material is moved along by a **number of jumps and bounces**, much how a pebble is moved along the bottom of a stream bed.

Wind Erosion

- The grains do not rise higher than about 1meter, and they move in the same direction the wind is blowing.
- Dust particles (silt and clay) can be carried along great distances and at greater heights than sand particles

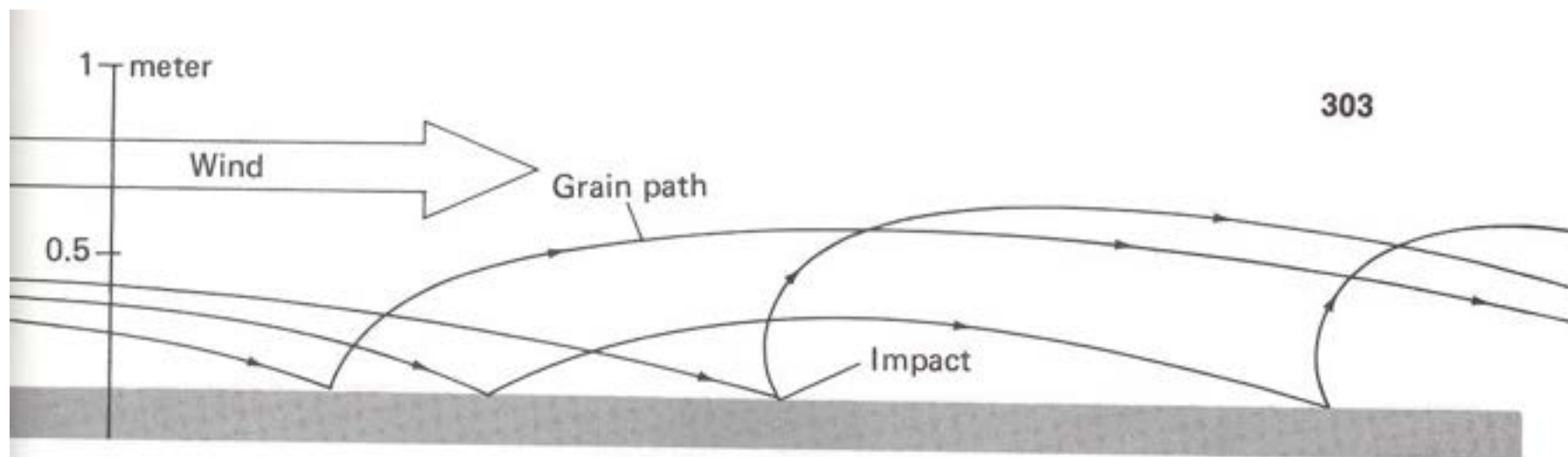
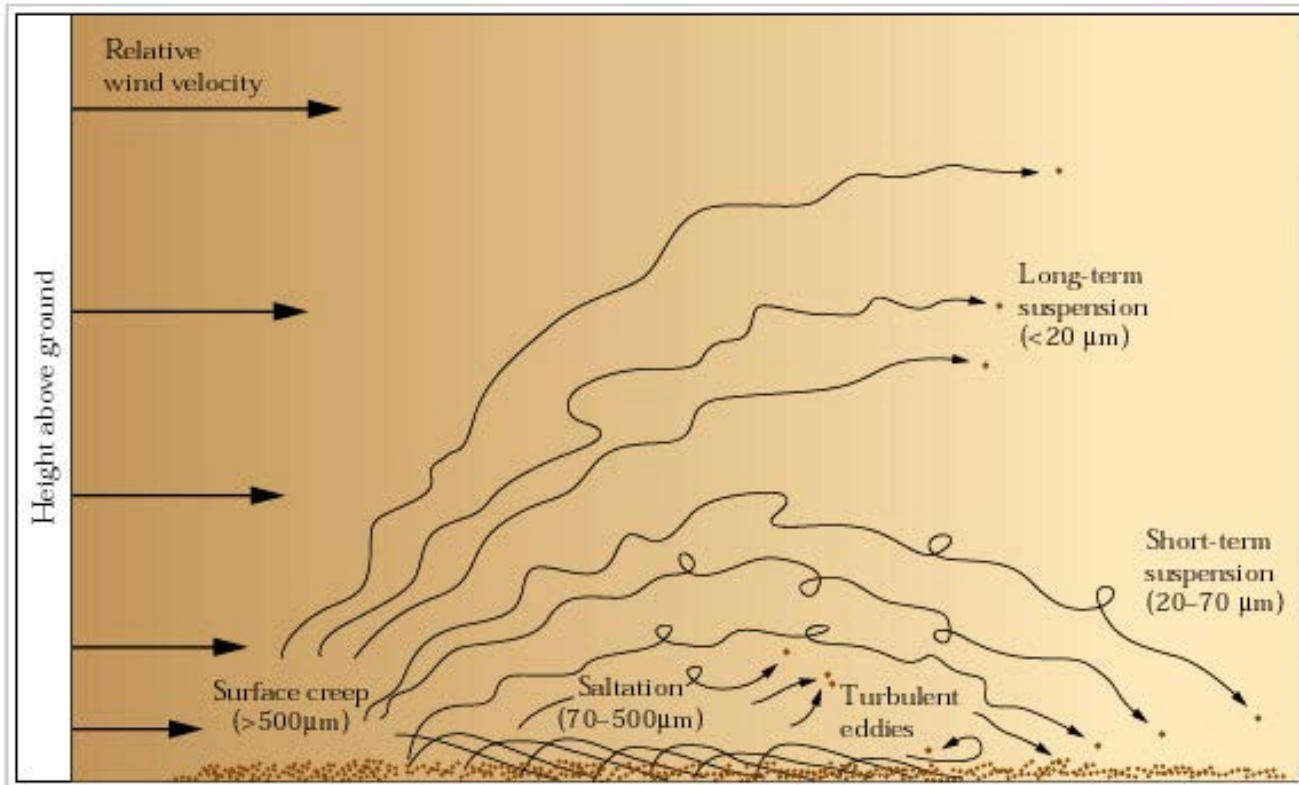


Figure 15-19. Wind erosion. Sand grains carried by the wind generally bounce along the ground. From hitting against other grains, the grains become rounded and their surface frosted.

Wind Transport



Moderate winds move the largest sand grains by surface creep. Slightly smaller sand grains move forward by saltation (bouncing). Finer particles are carried aloft where faster wind transports them downwind before they slowly settle to the ground. The very finest dust particles reach greater heights and are swept along in suspension as long as the wind keeps blowing.

Effects of wind erosion

- **Abrasion** : weathering of rock particles by the impact of other rock particles.
- In areas where there are strong, steady winds, large amounts of loose sand, and relatively soft rocks, abrasion causes a great amount of erosion.

Effects of wind erosion

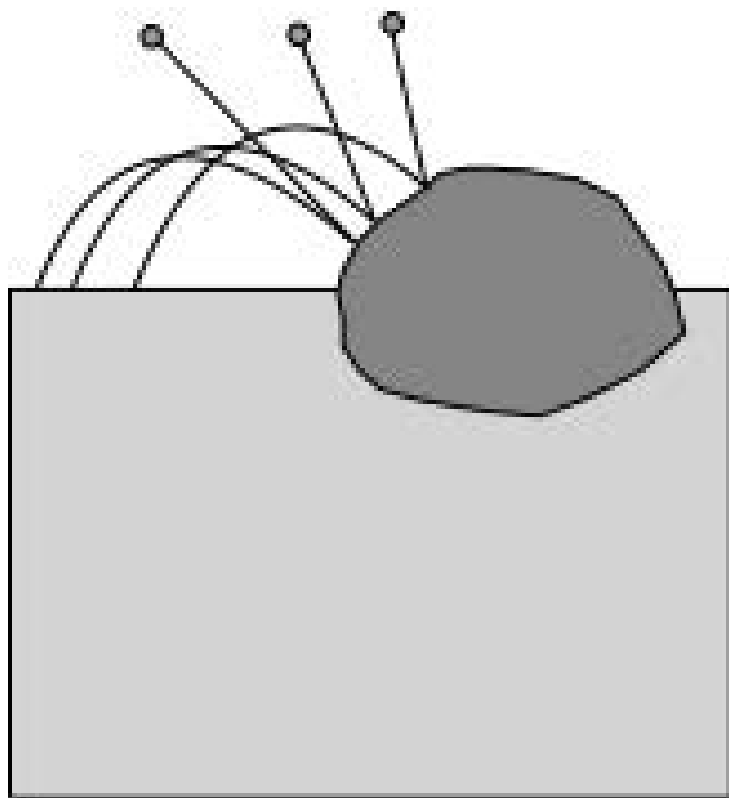
- Pebbles and small stones exposed to wind abrasion show surfaces that are flattened and polished on two or three sides.
- Rocks smoothed this way are called **ventifacts**.

Erosional Landforms

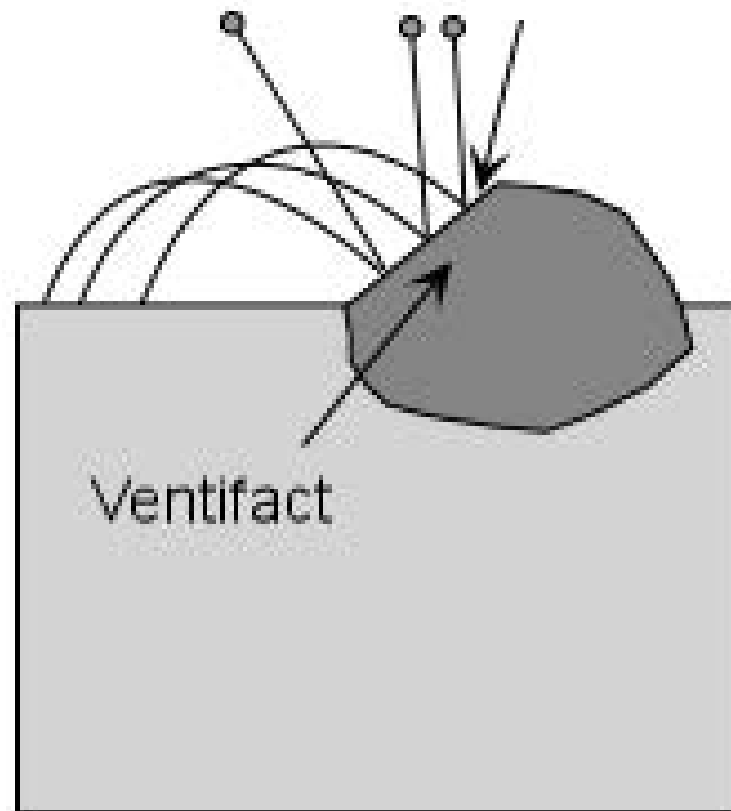
Ventifact

- Created by wind erosion
- Wind carries fine particles that work like a sand blaster (i.e. sand, silt, clay, and ice particles)
- The **windward face of the rock** is flattened and smoothed
- Usually pebble to cobble sized

Wind Direction



Abraded Surface



Ventifact



Dreikanter

- A type of ventifact that has three ridges
- Multiple faces have been formed by either a changing prevailing wind or movement of the actual rock being weathered

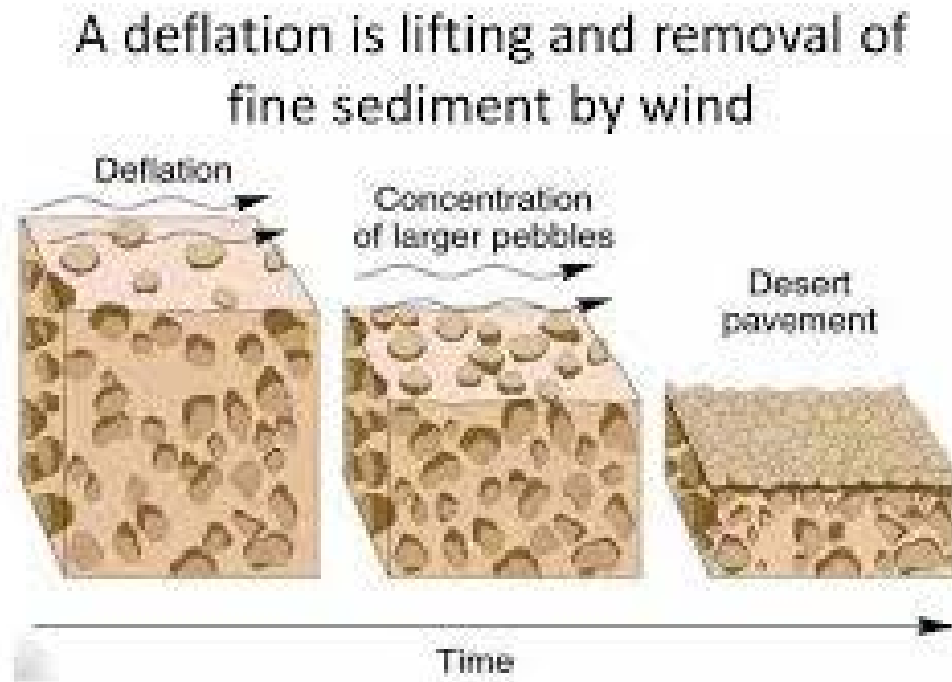


Desert pavement

- Formed in arid environments when wind carries finer, more lightweight particles such as sand away
- Large particles are left behind and protect from further erosion

Deflation

- Low spots carved out by wind erosion
- High elevations are left due to vegetative stabilization





Pedestal rock

- Base of a rock is weathered and eroded more quickly due to sand blasting
- Sand grains can only be picked up a couple of feet

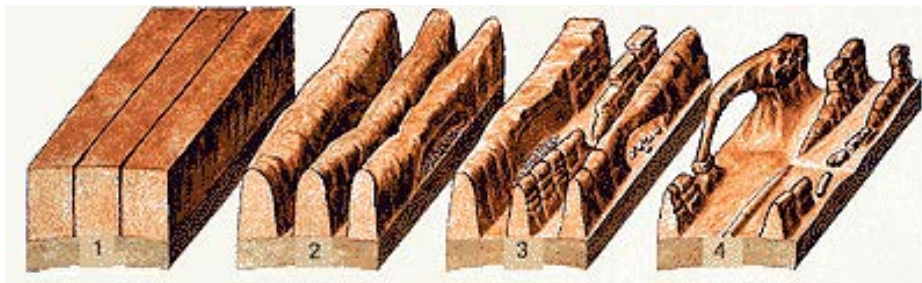
Figure 15-20. Erosion of a rock by wind. Because the particles transported by the wind remain close to the ground, the base of the rock undergoes the greatest amount of erosion.







Desert Arch



- Typically formed by wind
- Narrow ridges are formed and the softer substrata is weathered first, forming a bridge/arch overhead

Depositional Landforms

Loess

- Yellowish, fine grained silt and clay sized particles
- Carried and deposited by wind

Loess

- Criteria for Loess formation:
 - dust source (Periglacial and non-glacial deposits)
 - adequate wind energy to transport the dust,
 - a suitable accumulation area
 - sufficient amount of time

Loess



甘肃靖远山川景观

Sand Dunes

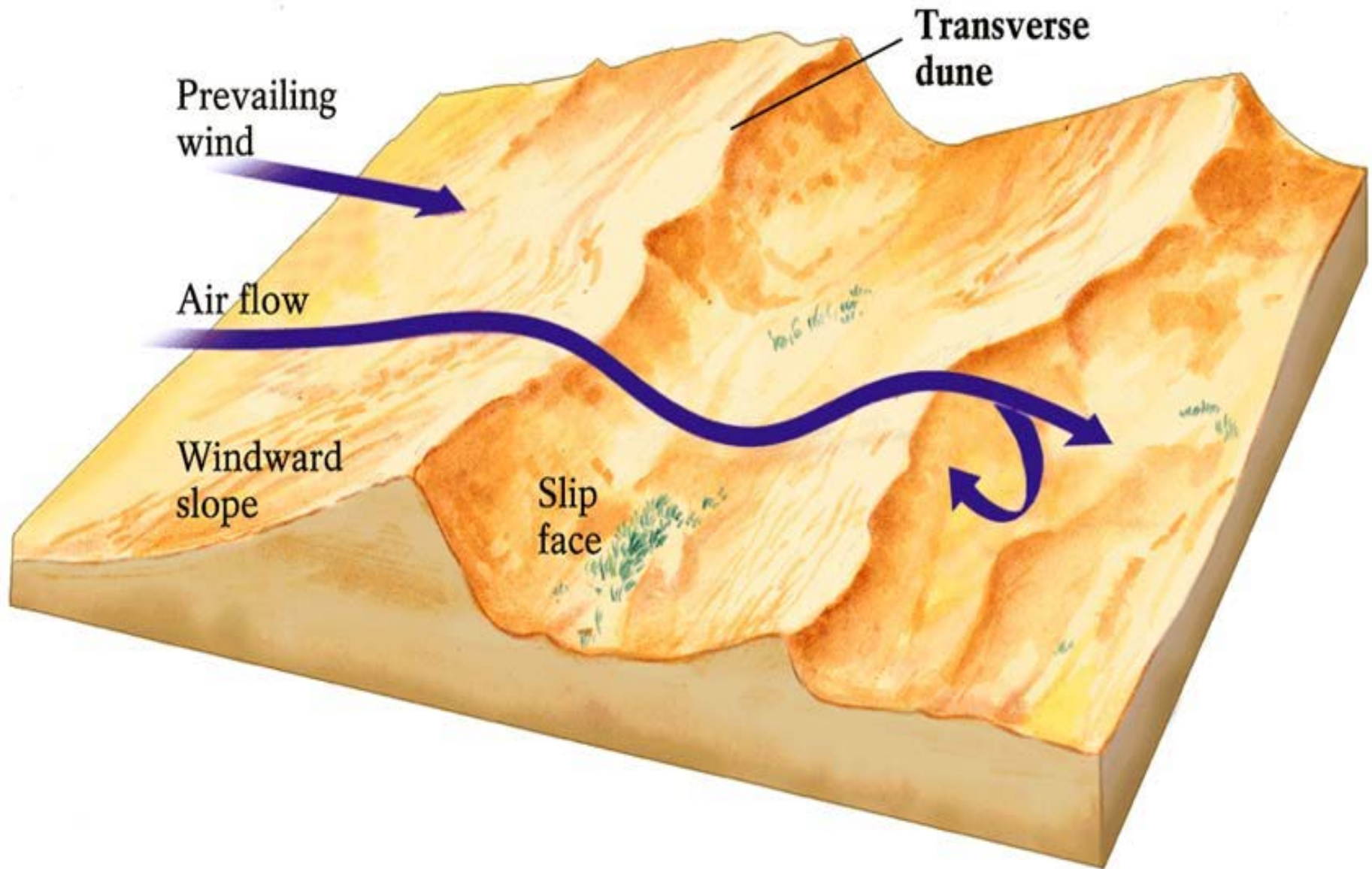
- Piles of sand
- Deposited by wind
- Leeward side (slipface) has a steeper slope
- Windward side is more gradual



Deposition and Dune Types

- Dune Types
 - Transverse- ridges that are perpendicular to prevailing wind direction
 - Longitudinal- ridges that are parallel to prevailing wind direction
 - Barchans- crescent-shaped with horns pointing downwind
 - Horseshoe (Parabolic)- crescent-shaped with horns pointing upwind
 - Star- winds from three or more directions

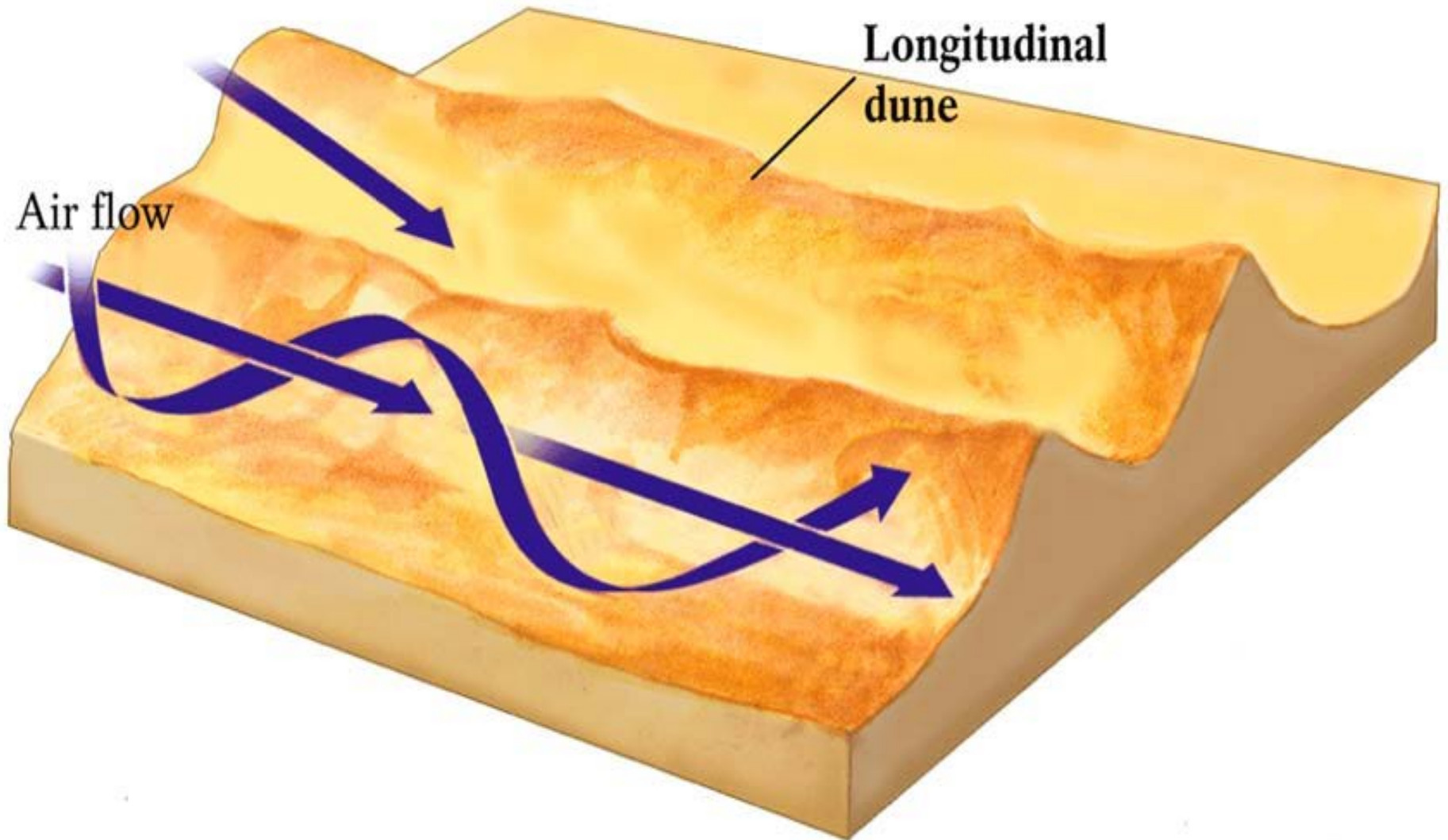
Transverse Dunes



Transverse Dunes



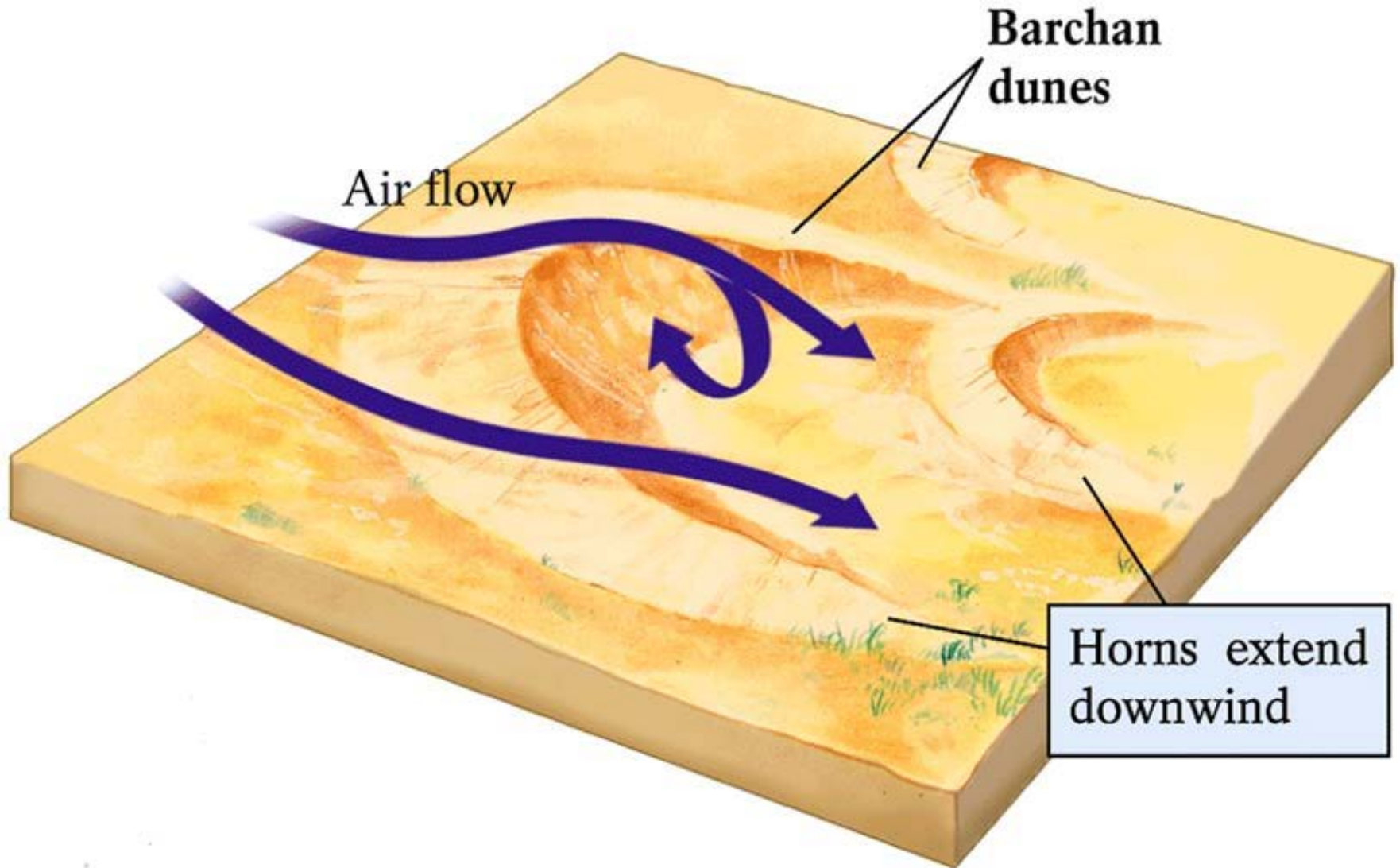
Longitudinal Dunes



Longitudinal Dunes



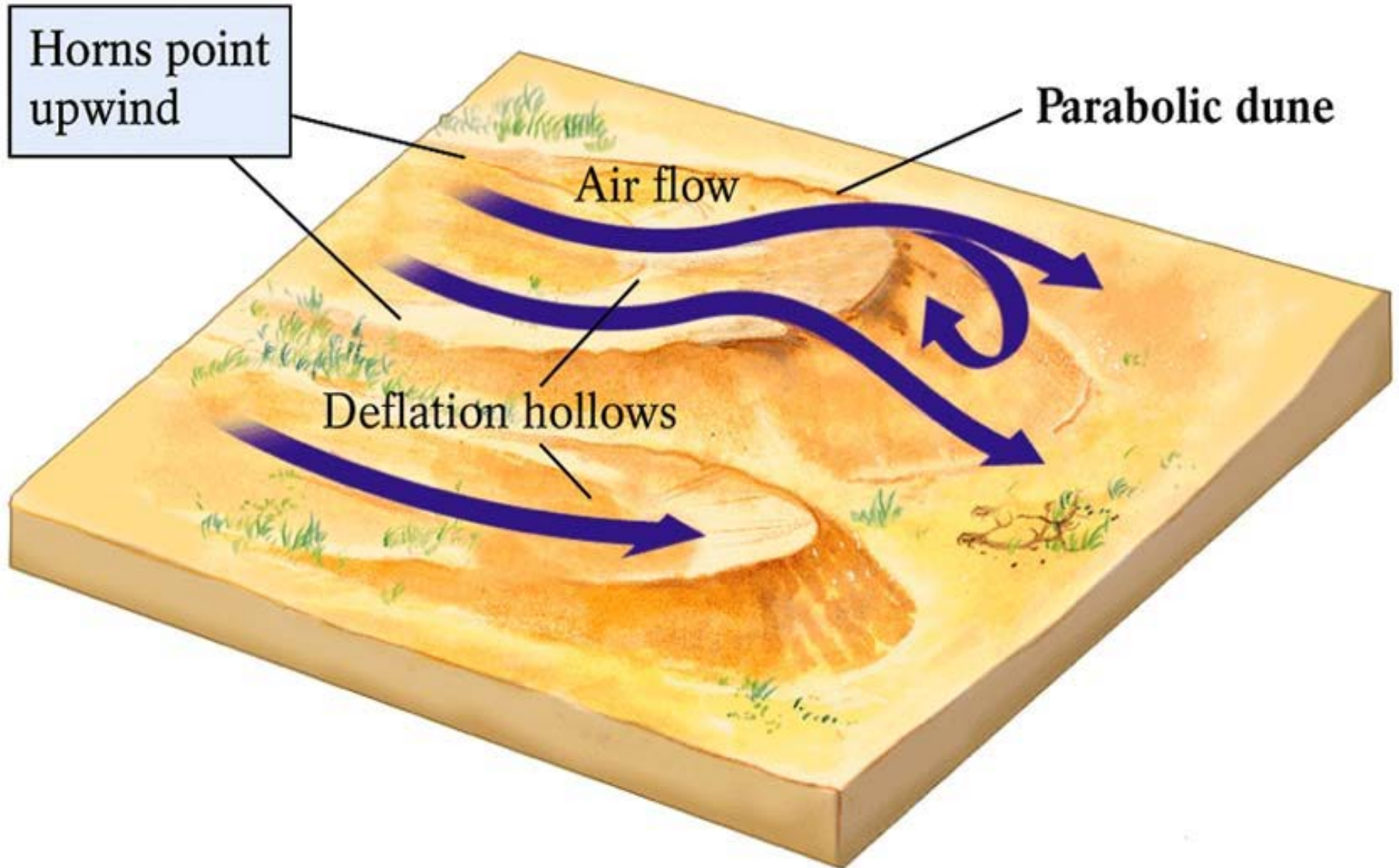
Barchan Dunes



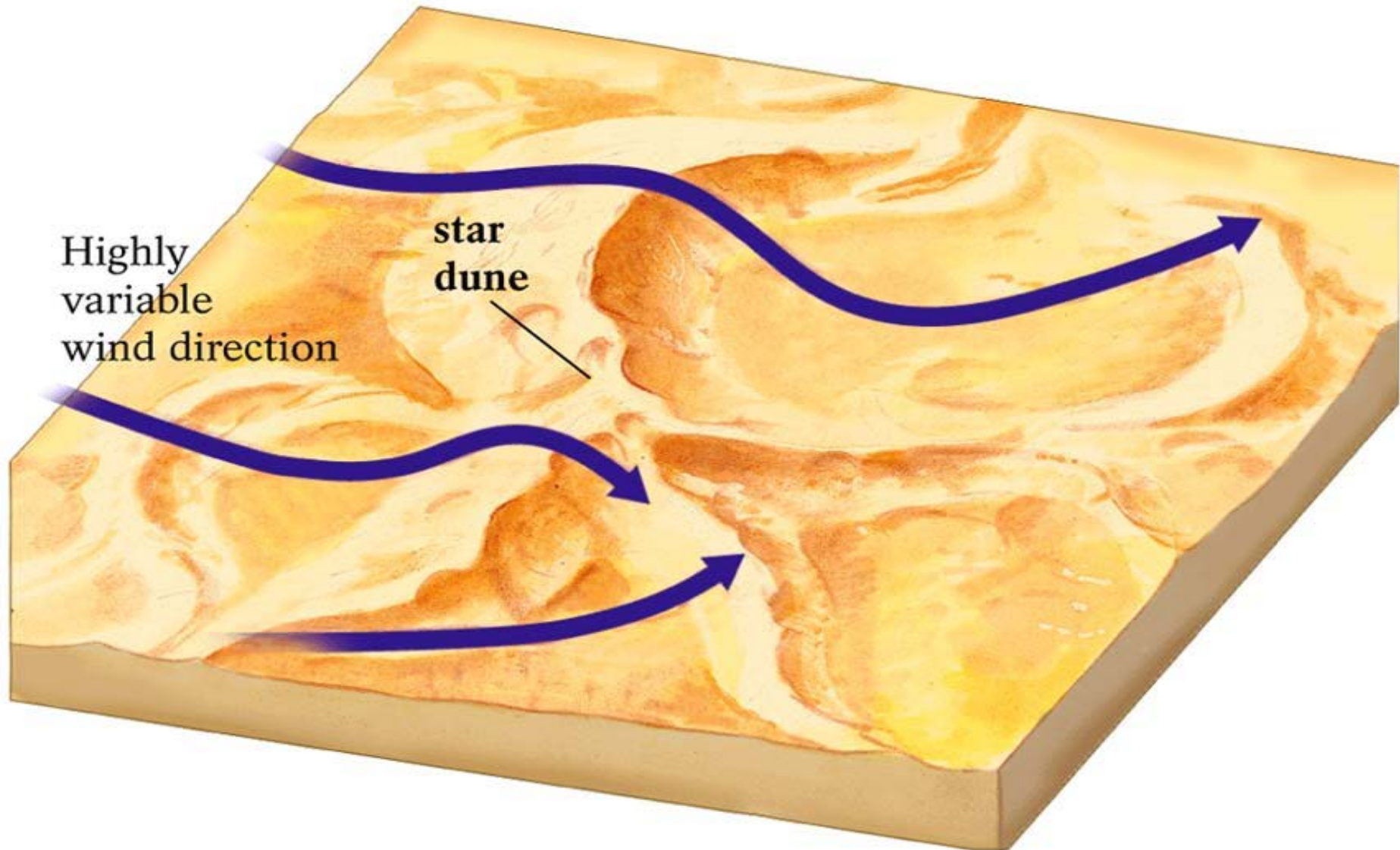
Barchan Dunes in Baja California



Parabolic Dunes



Star Dunes



Star Dunes



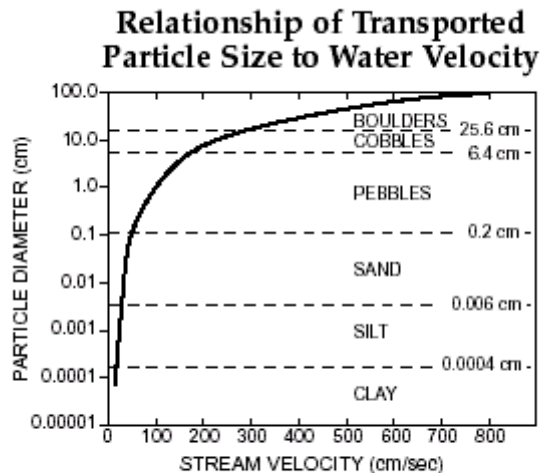
Ergs

- Huge dune fields
- Notice the source of sand from the photos



Streams as weathering/
depositional agents

Relationships in Streams



*This generalized graph shows the water velocity needed to maintain, but not start, movement. Variations occur due to differences in particle density and shape.

- The size of the sediments that a stream can transport increases as the stream velocity increases.
- The total amount of sediment that a stream can transport increases as its discharge increases.

Life History of a Stream

- **Youthful Stream**
- Steep gradients
- V-shaped cross sections
- Rough sediments flowing rapidly down stream.
- Due to greater water velocity larger sediment can be moved.

Life History of a Stream

- Sediments cut through bedrock as they are moved along
- Common to find rapids and waterfalls due to differences in resistance of the bedrock to weathering.



Mature Streams

- Potential energy for cutting and removing rock becomes less.
- The average gradient is decreased.
- Velocity near the bed becomes less.
- The size of sediment that can be moved decreases.
- Bed becomes covered with loose material, thus protecting it from further erosion.

- Cutting action of the stream becomes very slow.
- The base of the V has been widened due to weathering and the action of tributaries the valley now has gentle slopes.
- Meanders begin to form
- There is an increase in the volume of water that is carried.
- A greater mass of sediment can be carried, but most of it is silt and clay.



Old Streams

- Gradient becomes extremely small and only the finest of sediments can be moved.
- During times of peak flow the banks will overflow and flood the nearby portions of its valley.
- When the flow subsides a layer of silt and clay is left behind on the valley surface, this is the **flood plain**.

Meandering streams

- Meanders are strongly looped and from time to time they become cut off to form **oxbow lakes**.

Water Velocity in a meander

- Water velocity is greatest along the outside curve of a meander. Here erosion is dominant.
- Water velocity is slowest along the inside of a meander. Here deposition is dominant.

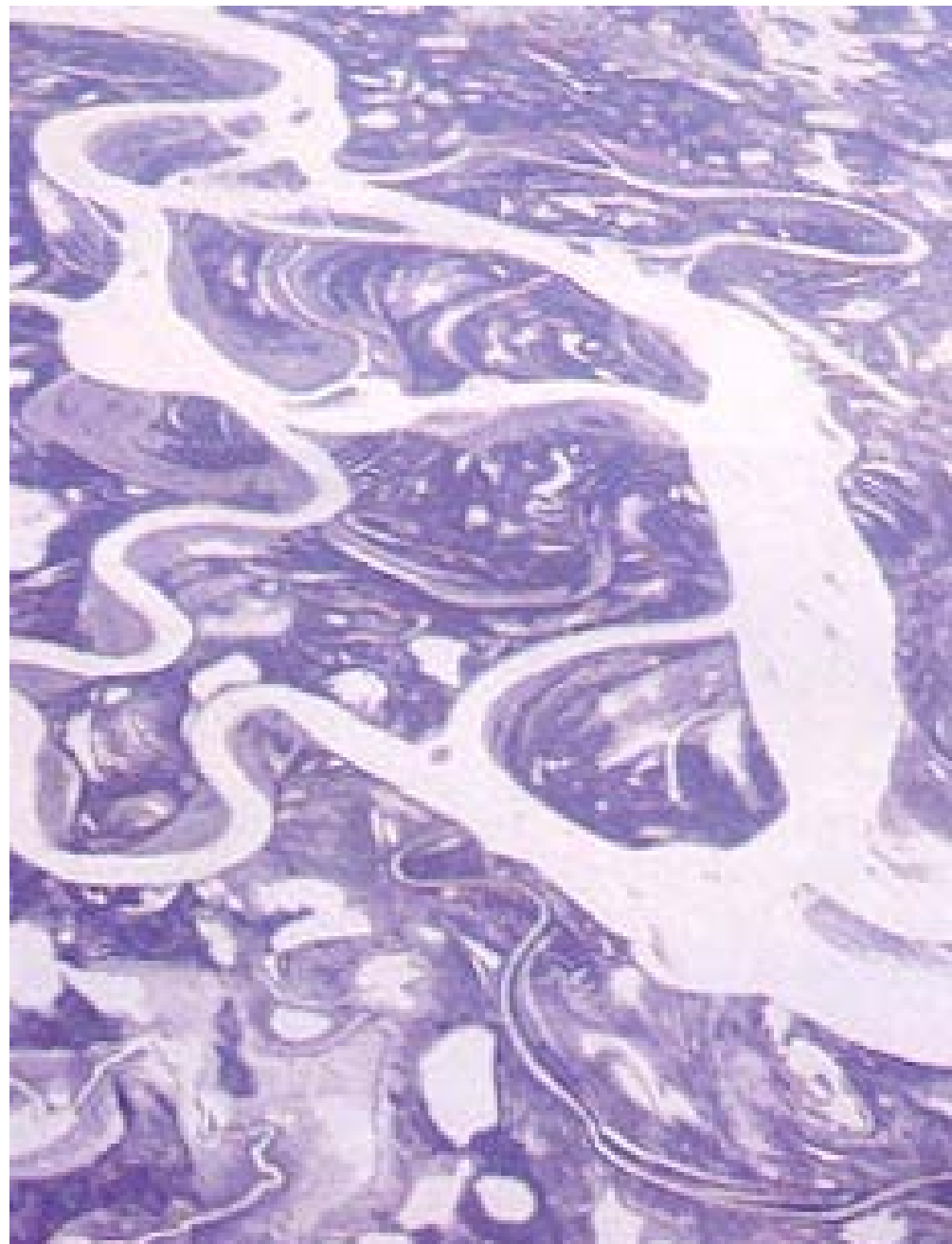
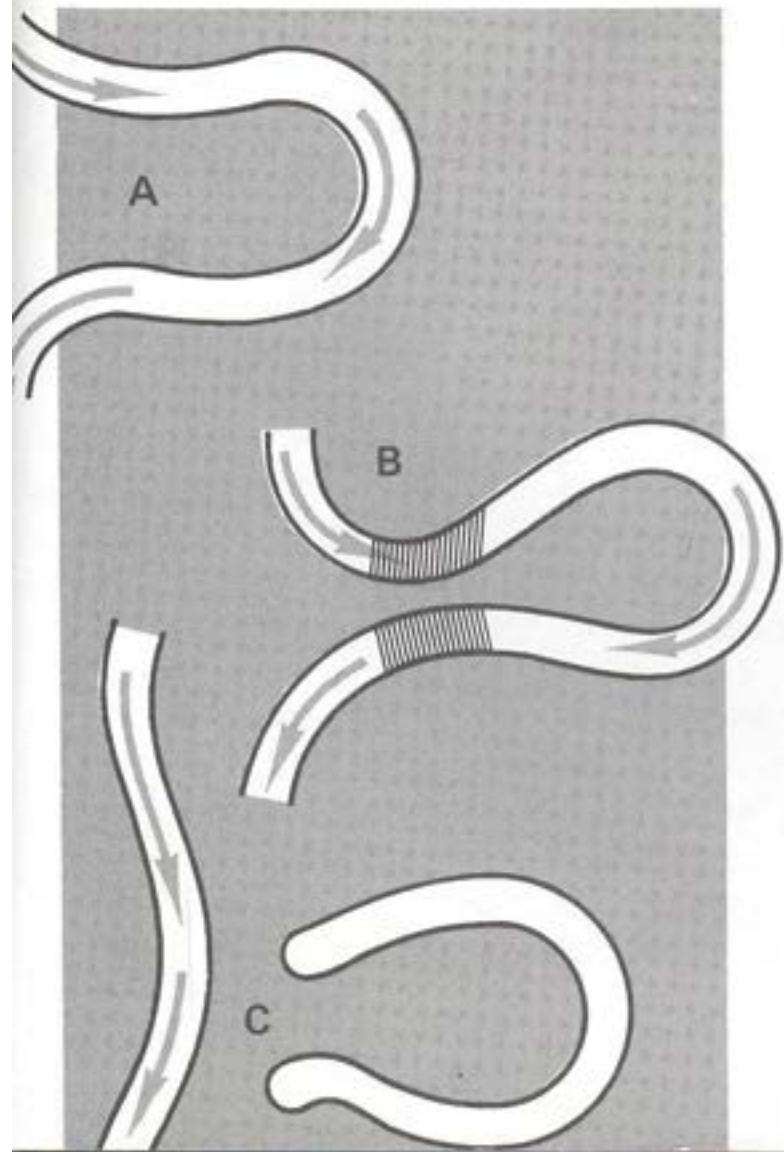




Figure 15-14. Stream in old age.

Figure 15-15. Formation of oxbow lakes.



Outside of meander

Inside of meander: deposition is dominant

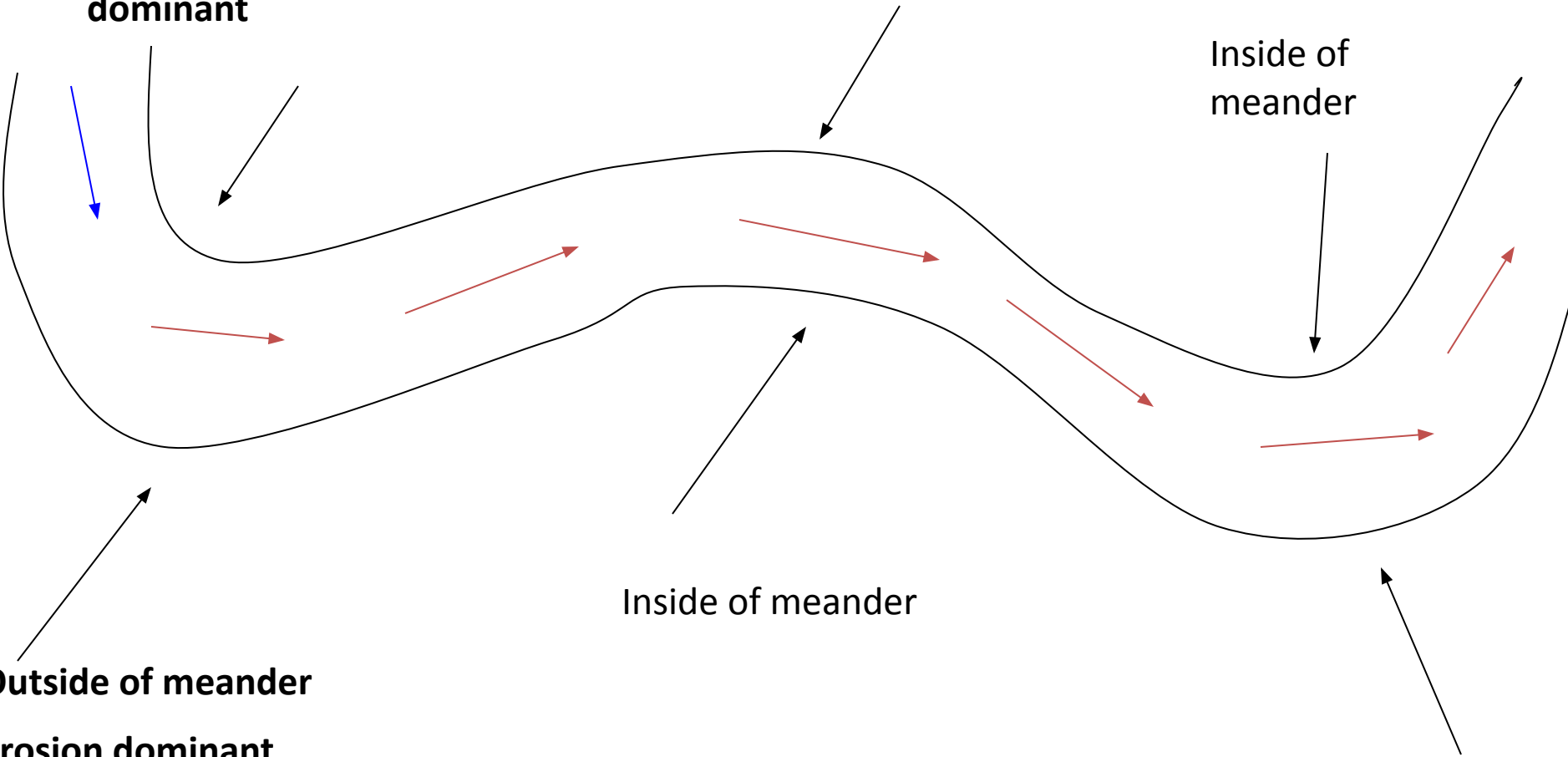
Inside of meander

Inside of meander

Outside of meander

Outside of meander

Erosion dominant



Locations of Stages

- It is unlikely that any stream is at the same stage of development throughout its entire length.
- Most streams tend to have the characteristics of youth near their source and of old age near their mouths and to be in the mature stage somewhere in-between..